JobConnect AI — Smart Job Recruitment & Career Platform

System Architecture Specification & Technical Documentation — Category: Full Stack & Applied AI

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1. Executive Summary & Problem Statement

Modern recruitment systems suffer from fragmented communications, manual resume screening bottlenecks, and lack of objective candidate-job matching. JobConnect AI addresses these challenges by delivering an integrated, enterprise-ready recruitment management platform serving three distinct user personas: Job Seekers, Recruiters/Employers, and Platform Administrators.

The platform unifies job discovery, resume creation, ATS scoring, candidate algorithmic ranking using TF-IDF cosine similarity, interview scheduling across 5 rounds, and downloadable PDF document generation.

2. Objectives & Scope

• Enable job seekers to create professional profiles, analyze resume ATS compatibility (0-100 score), and receive AI match percentages for active job postings.

• Provide recruiters with complete hiring pipelines, automated candidate scoring against job descriptions, and one-click offer letter generation.

• Equip administrators with governance consoles, company verification workflows, job listing moderation, and platform activity audit logs.

3. Technology Stack

Layer	Technology	Architectural Role
Frontend	React 18, Vite, Tailwind CSS	SPA client with role-based dashboard rendering & Axios interceptors
Backend API	Python 3.13, Django 5.2, DRF	RESTful API engine, SimpleJWT auth, role-based permission classes
AI / ML Engine	Scikit-learn, TF-IDF, NumPy	Vectorization of resumes and job descriptions with cosine similarity
Document Engine	ReportLab 5.x	Programmatic rendering of ATS resumes, receipts, and offer letters
Database	PostgreSQL / SQLite	Relational persistence with foreign-key integrity across users and jobs
Security	SimpleJWT, django-cors-headers	Bearer token authorization, CSRF protection, PBKDF2 hashing

4. System Architecture & Data Flow (DFD Level 0 & 1)

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[Client: React SPA] ■■(HTTPS/REST + JWT Bearer)■■> [Django REST Framework]
       ■                                                    ■
       ■■■ State Management (Axios + Context)                ■■■ SimpleJWT Auth Filter
       ■■■ Dynamic Role Dashboards (Seeker / Recruiter / Admin)  ■■■ Business Logic Controllers
                                                            ■   ■■■ Resume Analyzer
                                                            ■   ■■■ TF-IDF Match Engine
                                                            ■   ■■■ ReportLab PDF Generator
                                                            ■■■ Django ORM
                                                            ■■■ [PostgreSQL / SQLite Database]
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5. Core Database Schema & Relational Models

Table Name	Primary Key	Core Fields & Relational Foreign Keys
users_customuser	id (BigAuto)	email (unique), role (seeker/recruiter/admin), is_verified, date_joined
recruiter_company	id (BigAuto)	user_id (FK: users), company_name, website, industry, verification_status
jobs_joblisting	id (BigAuto)	recruiter_id (FK: users), title, description, skills_required, salary, status
resumes_resume	id (BigAuto)	user_id (FK: users), file_url, parsed_skills, experience_years, ats_score
applications_application	id (BigAuto)	job_id (FK: jobs), seeker_id (FK: users), match_score, status, stage
interviews_schedule	id (BigAuto)	application_id (FK: applications), round_type, scheduled_at, meeting_link

6. API Endpoints Architecture

• **POST /api/auth/register/ & /api/auth/token/**: User account creation and JWT token issuance.

• **GET /api/jobs/ & POST /api/jobs/create/**: Job search with filtering by work mode, location, and recruiter posting.

• **POST /api/resumes/analyze/**: Submits candidate resume for instant ATS lexical scoring and missing keyword analysis.

• **POST /api/applications/match/**: Computes TF-IDF cosine similarity between applicant resume tokens and job requirement vectors.

• **GET /api/documents/offer-letter/:id/pdf/**: Streams server-generated ReportLab PDF offer letter document.

7. Security, Testing & Limitations

• **Security:** Tokens stored in client-side secure memory/storage; DRF permissions enforce `IsAuthenticated` and custom role classes.

• **Testing:** Unit tests for TF-IDF tokenization sanity and API response status codes; database seeding script verified.

• **Limitations:** Currently relies on local/standard SQLite for development with production switchable to PostgreSQL via environment variables; third-party email notifications require external SMTP gateway configuration.

Architecture Note: Role-based access control (RBAC) ensures job seekers cannot inspect competitor applications, and recruiters can only manage postings associated with their verified organization.